Reducing a count can skip the exact target

The Mac processed the full, previously audited cohort of 1,024 destination-account/minute queries. The contrast is clearest in the **68 fully evaluated queries with three-event chains to reduce**.

68 of 68

Met the upper bound

All pairs preserved; at most 3 deletions

13 of 68

Reached the exact target

Same queries, target, preservation and budget

What was evaluated

Complete local context	Queries	With triples	Without triples
Empty; no profile requested	179	0	179
Unsupported graph structure	280	155	125
Eligible; above 32-event cap	43	9	34
Eligible; all requested profiles completed	522	68	454
Total	**1,024**	**232**	**792**

Every induced event in the preceding 24-hour local context was retained. The sample covers 1,023 distinct roots; queries can overlap. Of 232 queries with triples, 77 fit the required tree structure, and 68 were profiled (29.3% of all 232).

Exact versus upper targets on the same 68 queries

Minimum pairs preserved	Upper bound met within 3	Exact target met within 3	Exact target impossible at any budget
100%	68	13	55
95%	68	14	54
90%	68	15	53

For baseline triple count B_3, each policy uses the same integer target $U = \text{floor}(3B_3/4)$. The comparators are “at most U” and “exactly U.” Preservation rounds upward to a whole pair. The last column is determined from a completed profile; the nine size-capped cases are excluded.

Beyond the known pilot. The exact-target comparison was motivated by q0011, a previously inspected gap. Removing that query leaves 54 gaps in the other 67 positive, fully profiled queries at full preservation. They belong to the same previously audited cohort.

What the profile adds

One event may participate in several chains. With every root-ending pair preserved, a middle or final event in a surviving pair must stay. Deleting an upstream first event removes its triples together. A deletion can therefore move past a requested count even when it satisfies the upper bound.

Two observed outcomes that changed when preservation relaxed

Query and baseline (pairs, triples)	Full preservation	First looser tested threshold with an exact hit
q0204: (12, 6); U = 4	Cost 1 gives (12, 3). Exactly 4 is impossible.	Keep ≥11 pairs (90%): (11, 4) costs 1.
q0208: (20, 2); U = 1	Cost 1 gives (20, 0). Exactly 1 is impossible.	Keep ≥19 pairs (95%): (19, 1) costs 1.

These are the only two of the 68 for which relaxation changed exact-target attainability. Fifty queries had identical integer thresholds at all three percentages; 15 had two thresholds and three had three. The 89 distinct positive-query tables had preservation slack R = 0, 1 or 2.

What was checked

Completeness of the run. All 1,024 extractions completed; no queries are pending. The native run reported 563 completed tables out of 653 planned distinct query/quota tables. All 90 skips were recorded at the pre-solve event cap. A resource skip yields no optimization conclusion.

Verification on the Mac. All 935 returned profile witnesses were replayed. The local exhaustive checker ran on 528 of 563 completed tables and agreed on all 528. It skipped 35 above its 16-event limit. It agreed on 47 of the 55 exact gaps at full preservation. No mismatch or timeout was reported.

Verification of the upload. The archive passed its integrity check; run hashes match the released code and protocol. The 1,024 case rows and 18,432 policy rows reconcile. Baseline counts match the previous cohort audit query by query. The archive contains aggregates; validation here did not re-execute deletions on raw transactions.

How to use the result

Use this as a descriptive illustration alongside the theorem. The nine capped positive cases had a median of 91 events (12 among the 68 solved). Even if all nine had attainable exact targets, at least 55/77 eligible positive queries still had gaps at full preservation. The 155 structurally unsupported positive cases are outside that denominator.

The prior cohort was selected without labels or solver outcomes, but was previously analyzed. This design does not measure fraud detection, predictive accuracy, causal effects or out-of-sample performance. The warm-cache run took 222.24 seconds on macOS arm64 / Python 3.9.6; it supplies no comparative speed claim.

Sources. Returned BATCH_RESULTS.zip: PROTOCOL, BATCH_SUMMARY, case and goal records. Dataset: IBM HI-Small, version 8, 5,078,345 source rows; Altman et al., NeurIPS 2023.